

Alisson Matheus Silva

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EXPERIENCE

Machine Learning Engineer

Aug 2025 – Current

MercadoLibre Inc.

- Designed and trained computer vision models for large-scale image classification tasks using YOLO and Vision Transformers (ViT), improving detection accuracy across production pipelines.
- Conducted security inspections of ML pipelines, identifying and remediating vulnerabilities in model serving infrastructure and data ingestion workflows.
- Assessed and enforced data quality standards for training datasets, implementing automated validation checks to detect label noise, class imbalance, and distribution drift.
- Automated end-to-end model training pipelines using orchestration frameworks, reducing manual intervention and accelerating iteration cycles. Implemented security checks with MCPs.
- Developed and executed red team attacks against ML pipelines, including adversarial input generation, data poisoning simulations, and model extraction probes to evaluate robustness.

Data Science Consultant

Sep 2024 – Aug 2025

Management Solutions GMS

- Developed and optimized credit risk models, processing large datasets with PySpark, Databricks, and SAS to assess default probabilities for financial institutions.
- Built classification trees and regression models using scikit-learn and XGBoost, implemented deep learning models with PyTorch and TensorFlow, conducted exploratory data analysis and feature engineering to improve model accuracy and training data quality.
- Contributed to a LangChain RAG pipeline for document validation and LLM-based workflows for sentiment analysis and automated data-quality checks, enhancing forecasting accuracy.

Junior Data Analyst

Mar 2024 – Sep 2024

Seed Digital SA

- Analyzed data collected from digital sensors and built professional dashboards in Excel and Power BI to surface trends and generate actionable business insights.
- Executed SQL queries to extract, clean, and manipulate datasets, enabling accurate reporting and data-driven recommendations.
- Built and refined forecasting models using Facebook Prophet, applied outlier detection and exponential smoothing to improve predictive accuracy.

EDUCATION

University of São Paulo (USP)

Jul 2025 – Current

Doctor of Philosophy, Physics

São Paulo State University (UNESP)

Mar 2022 – Sep 2024

Master of Science, Physics

University of Brasília (UnB)

Aug 2017 – Dec 2021

Bachelor of Science, Physics

CERTIFICATIONS

AWS Machine Learning Specialty

Apr 2025

- Proficient in designing, training, and deploying ML models within the AWS ecosystem using SageMaker and related services.

AWS Cloud Practitioner

Dec 2024

- Skilled in core AWS services, cloud security, pricing models, and cost optimization, able to align AWS solutions with business objectives.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Spark (PySpark), R, Bash

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Numpy, Pandas

MLOps & Cloud: SageMaker, Databricks, MLOps pipelines, SAS, AWS

Other: LangChain, RAG systems, MCP, Power BI, Excel, Facebook Prophet

HONORS AND FELLOWSHIPS

1st Place, Big Data Hackathon – Forecast 2025

Oct 2025

- Issued by Big Data.

1st Place, Itaú Unibanco Hackathon – Batalha de Dados Quantum

Jan 2024

- Issued by Itaú Unibanco.

Scholarship, Master's Degree Program (PIBPG)

Mar 2022

- Issued by the National Council for Scientific and Technological Development (CNPq).

Honorable Mention, 27th Congress of Scientific Initiation at UnB

Aug 2021

- Issued by University of Brasília (UnB).

Scholarship, Scientific Initiation Program (PIBIC)

Mar 2021

- Issued by the National Council for Scientific and Technological Development (CNPq).

LINKS

GitHub: github.com/itsalissosilva

LinkedIn: linkedin.com/in/itsalissosilva

Website: itsalissosilva.github.io

Hugging Face: huggingface.co/itsalissosilva

ADDITIONAL

Languages: Portuguese (Native), English (C2), German (B2), Spanish (B2)

Volunteering: Tutor at the IV Summer School of ICTP-SAIFR for Young Physicists, taught fundamentals of physics and algorithms for scientific research.

Interests: Marathon running, language learning, digital drawing and illustration.

This CV was last updated on February 2026.